

Serotonin and Dopamine Mediated Behaviours
Following MDMA and Amphetamine Treatment in the Serotonin Reuptake Transporter
Knockout rat

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Abstract

The behavioural effects of MDMA and amphetamine administration result in distinct behavioural profiles in rats. While there are some similarities between the drugs such as increases in hyperactivity there are also clear differences such as MDMA elicited serotonin syndrome behaviours, demonstrating unique neural mechanisms affected. Using the drug discrimination and locomotor activity paradigms, this study investigates the behavioural patterns of AMPH and MDMA using the serotonin reuptake transporter knockout (SERT^{-/-}) rat compared with wildtype (SERT^{+/+}) rats. During drug discrimination testing SERT^{-/-} animals take longer to *acquire* discrimination but respond quicker during generalisation tests when they do, demonstrating that MDMA is comparable to AMPH in SERT^{-/-} animals. This is further supported by locomotor activity findings which show that SERT^{-/-} animals have a reduction in hyperactivity following MDMA treatment but increased hyperactivity following AMPH treatment, when compared to SERT^{+/+}. Measuring these differences is important for understanding the role 5HT play in the behavioural profiles of MDMA and AMPH.

(±)-3,4-methylenedioxy-methamphetamine (MDMA) is unlike classic psychostimulants because it elicits both stimulatory and hallucinogenic effects. Whereas classic psychostimulants typically have the highest affinity for the dopamine transporter (DAT), MDMA binds with a higher affinity to the serotonin transporter (SERT; Bankson and Cunningham, 2001), and with a 10-fold lower affinity to DAT and noradrenaline transporters (NET; Green, et al., 2003, Rothman, et al., 2001). MDMA binding not only blocks the SERT and DAT but also reverses the transporters resulting in an efflux of neurotransmitters from the vesicles, causing a rapid release of serotonin (5-HT; Green, et al., 2003; Concheiro, et al. 2013) and, to a lesser degree dopamine (DA) in the extracellular space (Green, et al.; Spanos & Yamamoto, 1989). Overall, the behavioural effects observed from MDMA administration are considered to be predominately due to combination of 5-HT and DA effects on the central and peripheral nervous system.

The neurochemical effects of MDMA result in physiological outcomes, which have been extensively investigated in humans and animals. In humans, the acute physiological effects include increases in heart rate, blood pressure, sweating, muscle tension, insomnia and dehydration (Green, et al., 2003) and some users experience jaw clenching or teeth grinding (Parrott, 2002). The excessive release of 5-HT induced by MDMA can lead to the so-called serotonin syndrome (SS). In humans, the SS is identified as hyperactivity, confusion, fever, tachycardia, tremors and hyperreflexia (Gillman, 1999). Mild SS is the combination of two or three of these symptoms and generally does not require medical attention (Parrott, 2002), however, a combination of most of the symptoms could be fatal if they are not attended too (Gillman, 1999). As such these symptoms and effects of MDMA identified in humans demonstrate the consequences of excessive 5HT amounts and the physiological action that excess precipitates.

Additionally, animal models have been used to investigate the behavioural effects of MDMA, as they allow control for factors such as drug history, desirability biases and environmental factors. Similarly to humans, the physiological effects of MDMA in animals are increases in locomotor activity (LMA; Gold and Koob, 1989) and behavioural displays of the SS (Spanos & Yamamoto, 1989). SS in animals is characterised by hyperthermia, head weaving, piloerection, fore-paw treading,

salivation, defecation and hyperactivity (Green, et al., 2003). Interestingly, Green, et al., has suggested that at least the hyperactivity resulting from MDMA administration is likely to be a combination of its effects on both 5-HT and DA. However, it is not clear to what extent other MDMA mediated effects are also mediated via the combined 5-HT/DA action. Moreover, it would be interesting to selectively ‘isolate’ the dopaminergic or serotonergic component of MDMA and investigate how the behavioural profile of MDMA changes under such conditions.

(±)-1-phenylpropan-2-amine (amphetamine; AMPH) is a psychostimulant and commonly known as “speed”, “dex”, as well as being an active ingredient in the pharmaceutical medication Adderall. Drugs like amphetamines (AMPH) are some of the most commonly consumed and abused compounds (Degenhardt, et al. 2013). AMPH is prescribed in New Zealand as a pharmaceutical treatment for Attention Deficit Hyperactivity-Impulsivity Disorder and in some cases narcolepsy. Following administration of AMPH, there are increases of DA, norepinephrine (NE) and 5HT (Rothman et al., 2001). Like with MDMA, AMPH both blocks the DAT, SERT and NET and reverses these, thereby promoting the efflux of neurotransmitters principally dopamine (Fischer & Cho, 1979). Furthermore, AMPH promotes the release of 5HT at a much lower level when compared to the DA release (Kuczenski, Segal, Cho & Melega, 1995). Compared to MDMA, AMPH produces both similar and unique physiological and behavioural effects.

In humans, AMPH increases blood pressure (Stoops, et al., 2004) and the subjective effects include increases in energy, arousal, elation and euphoria (Berman, Kuczenski, McCracken & London, 2009). White, Lott and de Wit (2006) examined the acute subjective effects of AMPH in a sample of healthy adult volunteers, using measures of personality traits and subjective responses. Specifically, White, Lott and de Wit, found that participants with low trait fear experienced stronger positive effects on a lower dose than those with high trait fear. The findings of the experiment demonstrated that the subjective experiences of AMPH can vary depending on the personality of the individual. Furthermore, the behavioural effects of AMPH are identified in animal models, particularly with regard to locomotor activity. It has been found that following the administration of AMPH, the resulting DA increases lead to hyper-locomotion (Pijnenburg, Honig, Van der Heyden &

Van Rossum, 1976). Antonious, Kafetzopoulos, Papadopoulou-Daifoti, Hyphantis and Marselos (1998) found that low doses of AMPH (0.5mg/kg) produce increased locomotion whereas at higher (3 and 6mg/kg) AMPH produces repetitive behaviours, such as grooming and head weaving. These repetitive and stereotyped behaviours following AMPH administration were investigated by Sloviter, Drust and Connor (1978). It was found that repetitive behaviours and displays of the SS were evident when rats had been treated with very high doses (20, 40, 80mg/kg) of AMPH. This provided behavioural evidence that AMPH also has a significant effect on 5HT-ergic system, although very high doses are required.

The physiological and behavioural effects of MDMA and AMPH show both similarities and differences demonstrating that they are mediated, at least in part by different neural mechanisms. One interesting way to investigate similarities between drugs is through the use of the drug discrimination paradigm. In this paradigm, animals are trained to discriminate the acute interoceptive effects of a particular dose of drug from vehicle through reinforced operant responses. During the test phase, the animal receives an untrained dose of drug. Responding is then contingent on whether the challenge *feels* more like the training drug or vehicle. Oberlander and Nichols (1988) used this procedure to demonstrate that rats could successfully discriminate between MDMA and vehicle (saline). Other studies have demonstrated that animals can discriminate between MDMA and other drugs of abuse, such as AMPH when trained on 1.5mg/kg of MDMA and tested with 0.5mg/kg of AMPH (Goodwin and Baker 2000). Similarly, Harper, Langen and Schenk (2014), demonstrated a dissociation between the stimulus effects MDMA and AMPH, when rats are trained on the same doses of MDMA at 1.5mg/kg and tested with AMPH at 0.5mg/kg. These findings indicate that the interoceptive cues were strong enough for the rats to discriminate between MDMA which has a high affinity for 5HT and a lesser effect on DA, and AMPH which has a high affinity for DA. Furthermore, the discrimination of MDMA and AMPH provides evidence for the different neurochemical mechanisms of MDMA and AMPH.

Whereas several of these drug discrimination studies suggest that the effects of MDMA are mediated by a combination of both enhanced 5HT and DA release, others have suggested that it is

purely mediated by 5HT or DA. Similarly, the repetitive behaviours and increased locomotor activity induced by MDMA have been thought to be mediated by a combined 5HT and DA effect (Green, et al. 2003). On the other hand the enhanced locomotor activity and stereotypy induced by AMPH has been thought to be exclusively mediated via its effect of dopamine neurotransmission (Antonious, et al. 1998; Sloviter, Drust & Connor, 1978). In order to investigate the relative contribution of the 5HT system to the behavioural effects of MDMA and amphetamine, we used a novel approach, making use of the serotonergic transporter knock-out rat. These rats were created using the ENU mutagenesis technique (Smits, et al. 2006), which induced a premature stop codon in the SERT gene. Homozygous SERT knockouts (SERT^{-/-}) completely lack the SERT and as a result have a 9-fold increase in basal extracellular levels of 5HT (Homberg, et al., 2007). This animal model provides a unique opportunity to examine the behavioural profile of MDMA. This is due to the lack of action on the SERT following MDMA administration, as there is no additional 5HT to be released into the extracellular space, therefore the drug cannot act on the SERT and does not lead to increased 5HT release in these animals. As such SERT^{-/-} is a genetic model of SERT deletion and provides an interesting avenue for investigating the role of 5HT in behaviour. This was illustrated by previous findings showing that fenfluramine (a drug which, like MDMA, enhanced 5HT release via an interaction with the SERT) was completely devoid of action in the SERT^{-/-} rat (Homberg et al 2007).

In addition, other behavioural changes have identified in the SERT^{-/-} rats. Olivier, et al. (2008) examined the baseline levels of anxiety in SERT^{-/-} in an open field test based on the measures of exploration. It was found that SERT^{-/-} spent less time in the centre of the arena when compared with wildtype litter mates (SERT^{+/+}) rats (Olivier, et al. 2008). To further support the findings from the open field test, Olivier, et al. examined duration and distance moved while in the elevated plus maze which has two open arms and two closed arms that are preferred by rats. The findings of Olivier, et al.'s investigation demonstrated that SERT^{-/-} spend less time in the open arms and move more in closed arms. Together these findings display that decreased time in the centre of open field as a measure of anxiety is a behavioural characteristic of the SERT^{-/-} rather than a reduction in exploration interests. In addition to Olivier, et al.'s investigation, the SERT^{-/-} was utilised in

Lizarranga, et al. (2014) study where following administration of 10mg/kg of MDMA, SERT^{-/-} had a delayed onset of hyperactivity and a decreased in vertical movements when compared with SERT^{+/+}. In all, the SERT^{-/-} model provides a valid prospect to examining behavioural changes when influenced by psychostimulants like MDMA or AMPH.

The aim of the present paper was to study the behavioural effects of MDMA and AMPH in both wildtype (SERT^{+/+}) and homozygous (SERT^{-/-}) rats. This was achieved using two different behavioural paradigms: drug discrimination and locomotor activity. In the previous literature discussed above, genetically unaltered animals have been used to examine the interoceptive cue induced by 1.5mg/kg MDMA and this dose seems to be partly mediated via dopamine and partly via 5HT₂ system. Thus, it is expected that a dose of 0.5mg/kg amphetamine (which is virtually devoid of 5-HT effects) would only partly generalize to the MDMA cue in SERT^{+/+} rats, whereas, in SERT^{-/-} rats, the same dose of MDMA can only affect the dopaminergic (DA-ergic) system; thus a more complete generalization. In addition, as the dopaminergic cue of 1.5mg/kg MDMA is much weaker than the 5HT cue, it is expected that the SERT^{+/+} will acquire drug discrimination faster than SERT^{-/-}.

With respect to locomotor activity, most studies have typically used a behavioural pattern monitor (BPM) which contains infrared photo beams distributed throughout a testing chamber, with the interruption of beam taken as an indication of movement (Flicker & Geyer, 1982). Although this allows for measures of total distance, velocity and rearing it is difficult to measure repeated movements like those seen in SS. Spanos and Yamamoto (1989) used a combination of manually coded behaviours of SS (head weaving and low body posture) as well as the BPM to analyse the increases in locomotor activity following MDMA administration. The results demonstrated that the administration of MDMA lead to dose-dependent increases of locomotion and SS behaviours when compared to behaviours following saline administration. Similar to Spanos and Yamamoto, these finer-grained analyses have been used to investigate other commonly repeated behaviours, like grooming along with SS behaviours. One such example is Antonious, Kafetzopoulos, Papadopoulou-Daifoti, Hyphantis and Marselos (1998) which combined the BPM analysis with an observation of specific animal behaviours such as rearing and grooming, to investigate the behavioural effects

following the administration of stimulants. The findings from Antonious, et al (1998) demonstrated that the typical methods for identifying locomotor show the dose response increases. However, Antonious, et al. specified that much more information can be gained from detailed analysis; particularly, when observed behaviours changing according to the dose administered to the animal. It is through the utilisation of different methods and animal models, that the physiological and behavioural effects of different manipulations can be analysed.

In the present paper we use a video-tracking system (EthoVision XT) designed to provide a very detailed analysis of movement by tracking three different body points (the tip of the nose, the centre of the body and the base of the tail). Additionally, we created a second observer- scored profile to obtain an even greater degree of detail. Moreover, this combination allowed us to compare the video-tracking data with the manual scoring to study whether the parameters of EthoVision can be used as a surrogate measure of manually scored behavioural categories such as grooming, pivoting or head-weaving. Given the above mentioned suggestion that the MDMA-induced hyperactivity is mediated (in part) by 5HT, it is expected that indices of SS (such as head-weaving) will be significantly lower in SERT^{-/-} rats after MDMA administration. Furthermore, given the suggestion that MDMA-induced hyperactivity is mediated by 5-HT, then the expectation is that the SERT^{-/-} will display reduced hyperactivity. Whereas, given the hyperactivity induced by AMPH appears to be predominately mediated by DA, it is expected that the AMPH induced hyperactivity in the SERT^{-/-} will not be reduced when compared to SERT^{+/+} rats. Finally, it is predicted that there will be dose dependent increases in activity for both genotypes following the administration of MDMA and AMPH.

Method

Subjects

A total of 49 male Wistar, 24 homozygous SERT knockout (SERT^{-/-}) and 25 wildtype (SERT^{+/+}) rats were used. Some animals were previously exposed to different behavioural paradigms but all were drug naive at the beginning of the current experiments. Animals ranged in weight 315g to 540g when testing began. Animals were group housed in polycarbonate cages (2-3/cage) in a temperature (19-21°C) and humidity controlled (55-60%) environment. All animals were bred in the VUW vivarium. All procedures were approved by the VUW Animal Ethics committee.

Animals used for drug discrimination (n=14; 8 SERT^{-/-}) were housed on reversed night/day cycle (lights on at 1900h). Animals were weight controlled to maintain 80-90% of their free feeding weight but had access to water *ad libitum*. Animals used for Locomotor Activity (LMA) (n=35) were kept on a day/night cycle (lights on at 0700) with free access to food and water. These animals were allocated into two separate groups, 19 (8 SERT^{-/-}) animals received MDMA and another group of 16 (8 SERT^{-/-}) animals received amphetamine.

Apparatus

Drug Discrimination

All training and testing occurred in 13 operant chambers (Medical Associates, ENV-221-M) between 1300 and 1800, 5 days per week during the animal's dark cycle. The operant chamber is equipped with two retractable levers paired with either a standard 100mA white or red light above each of the levers. Retractable levers were located to the left and right of a sugar pellet delivery device. MED-PC-IV (Med Associates) instrumentation and software was used to run experimental trials and for data collection.

Locomotor activity

All trials occurred in a circular open field (OF), which is 80cm in diameter with a 45cm high surrounding wall between 1300 and 1900. Activity was recorded via a camera suspended above the

open field at a distance. EthoVision XT v9 (Noldus) software was used for video tracking and data collection.

Procedure

Drug Discrimination

Auto Shaping. Training sessions initially consisted of 30 minute auto shaping trials until reliable responding was achieved, where the animals were consistently pressing on any presented lever to reach the fixed ratio (FR) allocated in the session. Rats started on a FR1 schedule which was increased to FR5, then FR10 before subsequent testing was conducted. This phase continued for 25 days.

Errorless training. During discrimination training trials, the session length was increased to 45 minutes and each animal received an intraperitoneal injection of MDMA (1.5 mg/kg) or vehicle (saline) immediately prior to being placed in the operant chamber. During errorless training trials only the drug (MDMA or vehicle) paired lever was presented, and responding was maintained on a FR10 schedule reinforced by delivery of a single sugar pellet. Following a 15 minute latency period, the paired lever is presented and the stimulus light is illuminated cueing the availability of the sugar reinforcer. The session ended when 60 reinforcers were delivered or 30 minutes had passed from the presentation of the lever. The dosing regimen was conducted in a pseudo-random order with no substance being repeated for more than two days consecutively. All rats completed 15 sessions of errorless training.

Discrimination. Following errorless training, all animals began 45 minute discrimination training sessions during which both levers were presented and sugar pellet delivery was contingent on responses on the correct drug-paired lever. Again, a 15 minute habituation period was used before the presentation of the levers (to allow the drug to take effect). This phase continued until the animal had learned to discriminate between MDMA and vehicle during the first 10 responses in a trial. An animal was considered acquired if 85% of responses during the first trial of a session were on the correct drug

paired lever for at least three consecutive days. Animals that did not meet this criterion within 55 training sessions were removed from the study.

Generalisation testing. Once an animal had met the criterion the animal was given a challenge injection of 0.5 mg/kg amphetamine (IP) immediately prior to the following session. Generalisation testing was identical to discrimination trials, except that sessions were terminated following 10 lever responses (i.e. achieving FR10 a single time) or if 30 minutes had lapsed. No reinforcement was delivered during the generalisation trial. A maximum of 2 generalisation sessions occurred in a week with at least one discrimination day between sessions for the animal to maintain 85% accuracy between testing. From the generalisation trials, measures of response time to reach the FR10 and the lever responses were recorded.

LMA

Animals were moved in their home cages to the experimental room and received at least 30 minutes habituation to the environment before testing. MDMA (2.5, 5.0, 10.0 mg/kg, SC) or AMPH (0.25, 0.5, 1.0 mg/kg, SC) was administered immediately prior to the animal being placed in the experimental chamber for 60 minutes. The doses used were counterbalanced in each drug condition except in the first session when all animals received vehicle (saline). All sessions were recorded using EthoVision XT (Noldus, Holland) software which tracks the animal's behaviour by monitoring three body points: nose point (NP; anterior tip of the nose), centre point (CP; centre of the back) and tailbase (TB; the proximal point of the tail). In addition to recording overall distance travelled and velocity, several other aspects of movement such as turn angle, meandering and body posture) were recorded for each of these three body points. A full list behaviours analysed can be seen in Appendix one.

In addition several behaviours were recorded manually for the first 30 minutes for the first 30 minutes of the session, including instances of grooming, rearing, head weaving and pivoting. For the MDMA trials, one experimenter coded the 0.0 and 10.0mg/kg trials and another experimenter coded 2.5 and 5.0mg/kg trials. The assessment of the interrater coding revealed no significant differences,

thus the manual scoring in the MDMA is equivalent across all trials. In consideration of the hyperthermic and stimulant effects of MDMA and AMPH, animals were singly housed overnight before being moved back to group housing. Test sessions were followed by a seven day washout period between doses.

Drugs

MDMA ($\pm$ 3,4-methylenedioxymethamphetamine hydrochloride; BDG Synthesis, Porirua) and Amphetamine (d-amphetamine sulphate; Sigma-Aldrich, Auckland) were dissolved in 0.9% sterilised saline and injected at 1ml/kg. Weights of the specified drug refer to the salt weight.

Data

All analyses were conducted using SPSS (IBM, v20). For the drug discrimination experiments differences between genotypes in acquisition were assessed using a Kaplan-Meier survival curve. A 2(genotype) x 2(lever) factorial ANOVA was used to measure the effect of genotype on number of responses during generalisation trials. Additionally, paired sample t-tests were conducted to evaluate both response time and drug-paired lever preference during the AMPH generalisation tests.

Data gathered from the LMA experiments was subjected to Pearson's correlations and a 2(genotype) x 4(dose) factorial multivariate ANOVA, followed up with post hoc comparisons of dose effects.

Results

Drug Discrimination

Kaplan-Meier survival analysis shows a significant leftward and upward shift in the survival curve of the SERT^{+/+} group compared with the SERT^{-/-} group ($\chi^2_1=5.248$, $p=0.024$; fig.1).

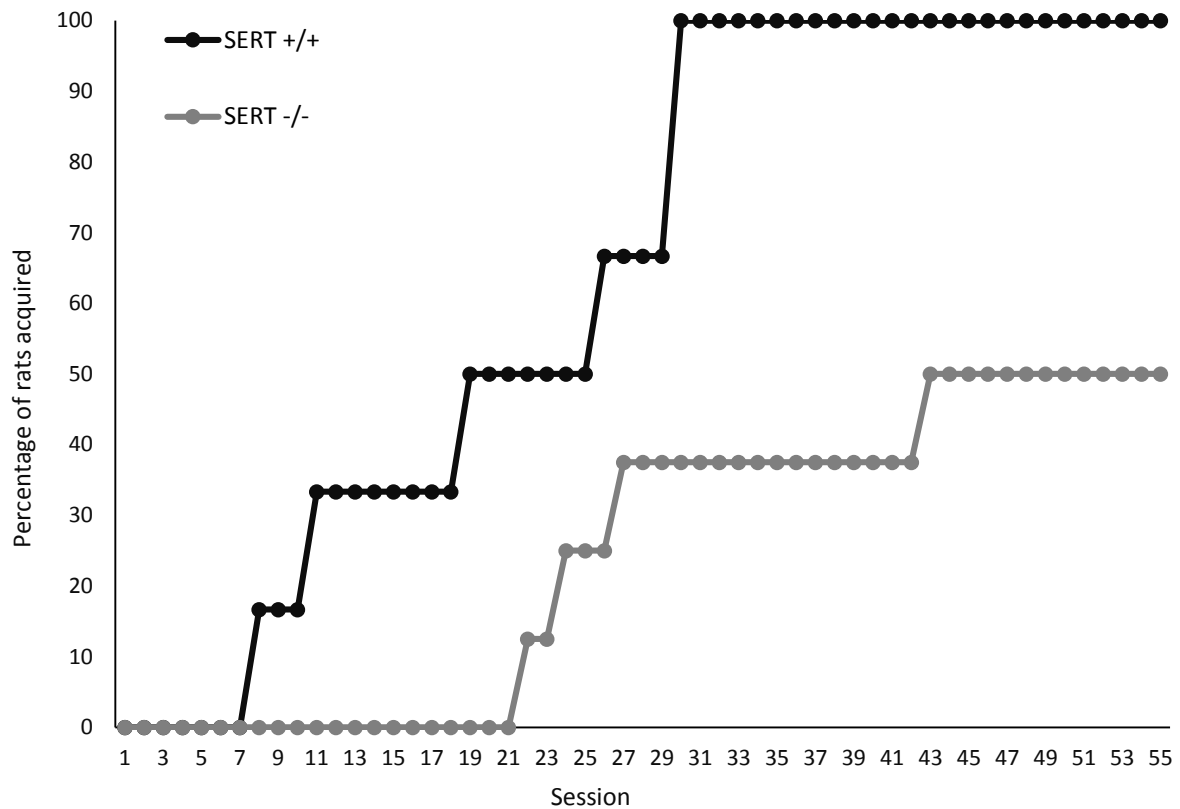


Figure 1. Survival curve showing the number of sessions to reach acquisition in the drug discrimination paradigm for SERT^{+/+} and SERT^{-/-} animals. N/n = 14/10.

Once the animals had reached acquisition, animals completed a generalisation test with 0.5mg/kg of AMPH. SERT^{-/-} animals had a significantly shorter response time to reach FR10 ($t(9) = 4.67$, $p=.001$, 95% CI [120.61, 347.65], $d = -2.07$; fig.2).

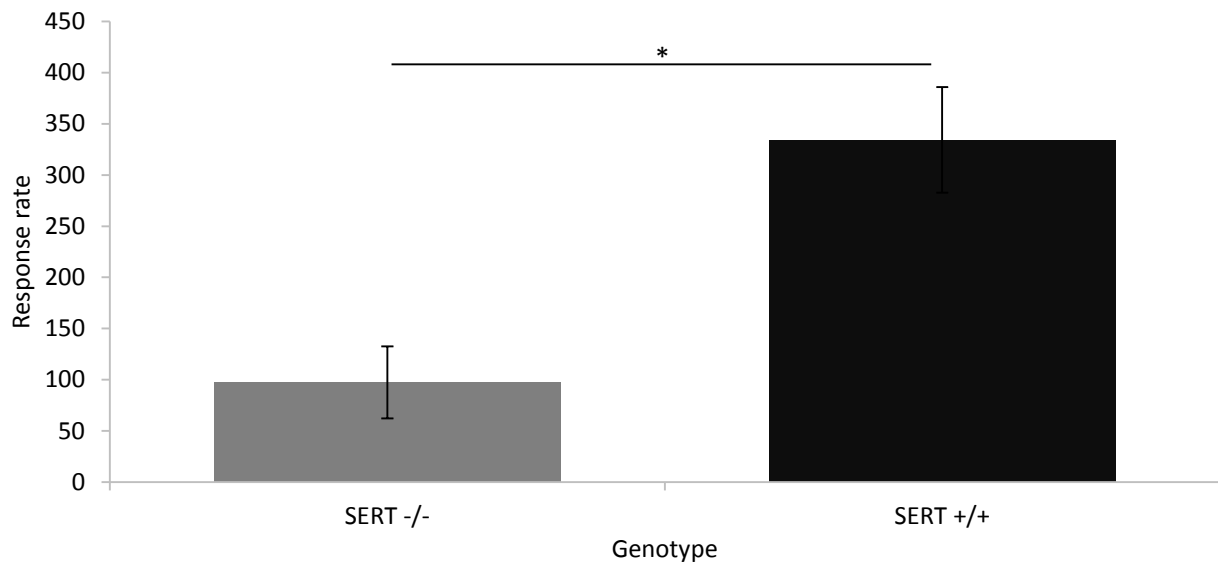


Figure 2. Mean latency to FR10 during AMPH generalisation tests. Error bars represent standard error of the mean. * = $p < .05$

Animals completed an average of 2-3 generalisation tests, the analysis of lever preference revealed a non-significant result in the Repeated Measures ANOVA (for lever preference ($F(1, 8) = .078, p = .787, n.s$) and for the genotype by lever interaction, ($F(1,8) = .312, p = .592, n.s$ as per fig. 3).

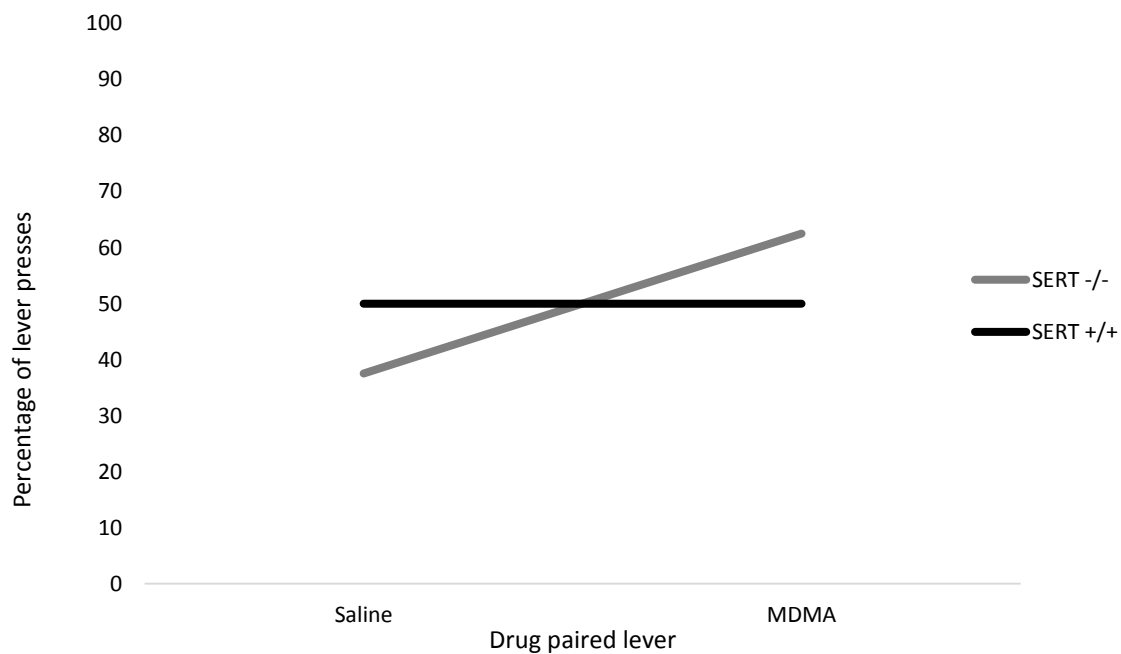


Figure 3. Mean percentage of lever presses on drug paired lever in generalization trials, in SERT^{-/-} and SERT^{+/+}.

Locomotor Activity

Analysis of data obtained from manually coding the first 30 minutes

Locomotor Activity following MDMA administration

The statistical analysis of the manually coded behaviour of grooming showed a main effect of dose ($F(3, 65) = 11.833, p = .001, \eta^2 = .353$; see fig. 4). To investigate the dose effect further, the post hoc comparison showed the dose of 0.0mg/kg was significantly different from 5.0mg/kg ($p < .05$), 10.0mg/kg ($p < .001$) and 5.0mg/kg was significantly different from 2.5mg/kg ($p < .001$). However, there were no effects of genotype or genotype x dose interaction were found.

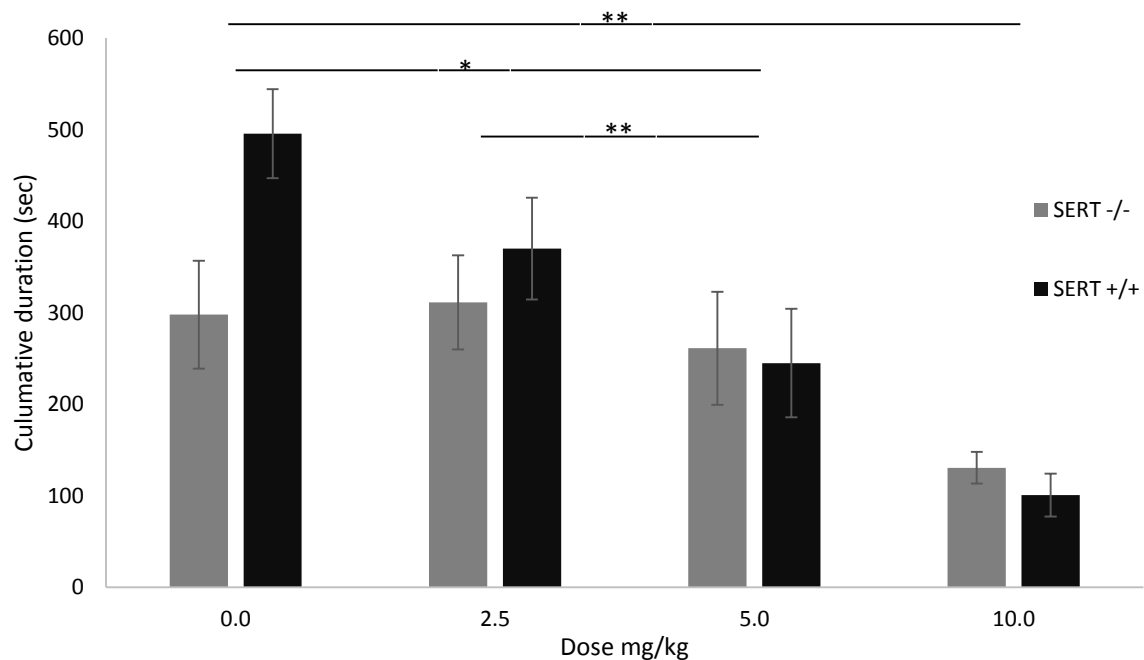


Figure 4. Mean duration of grooming by SERT^{-/-} and SERT^{+/+} for 30 minutes following administration of MDMA. Error bars represent standard error of the mean. * = $p < .05$, ** = $p < .01$.

A main effect of dose was found for both the duration of head weaving ($F(3, 65) = 3.002, p < .05, \eta^2 = .122$) and the mean count of head weaving ($F(3, 65) = 7.480, p = .001, \eta^2 = .257$). Both groups of animals spent comparable periods of time head weaving regardless of dose however there was a dose-dependent effect in the number of instance occurring with the session between 0.0mg/kg, 2.5mg/kg and 5.0mg/kg ($p < .05$), and between 2.5mg/kg, 5.0mg/kg and 10mg/kg and 5.0mg/kg ($p < .05$; as per fig. 5). The lack of difference between 0.0mg/kg and 10mg/kg suggests that there is

variance of this behaviour within this group of animals. Additionally, there were no other effects of genotype or interactions.

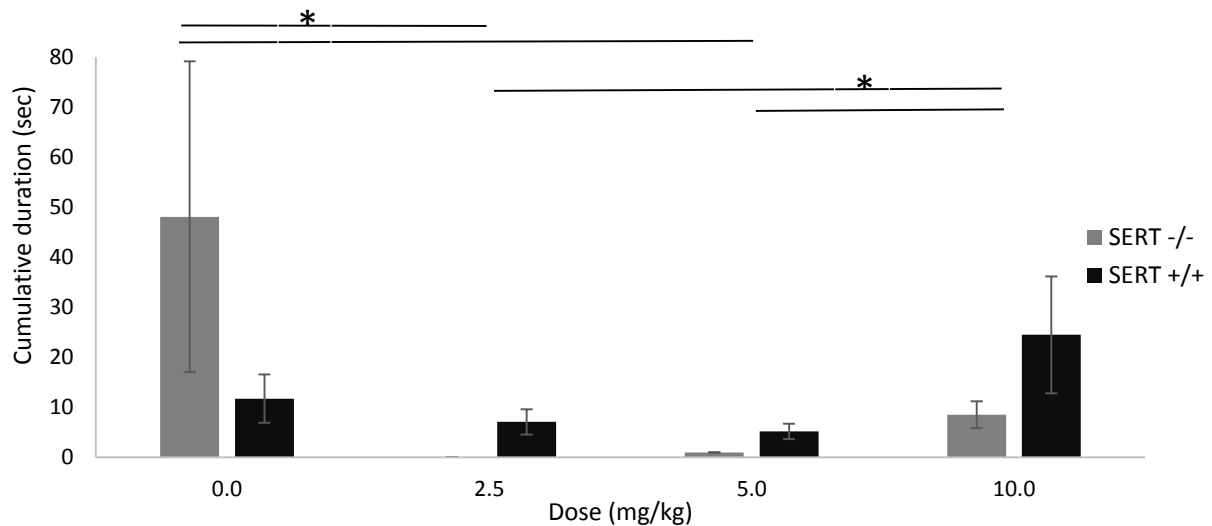


Figure 5. Mean duration in seconds the manually coded behaviour of head weaving in SERT^{-/-} and SERT^{+/+} following MDMA dose of 0.0, 2.5, 5.0, 10.0 mg/kg. * = $p < .05$.

Investigating the differences in the pivoting behaviour of genotypes on the four doses, there was a main effect of dose for the mean counts of pivoting ($F(3, 65) = 10.193, p < .001, \eta^2 = .320$) due to a significant difference between 0.0mg/kg and 5.0mg/kg ($p < .05$) and 10.0mg/kg ($p = .001$), and between 2.5mg/kg and 10mg/kg ($p < .05$; as per fig. 6). However, the genotype and interaction analyses revealed no other significant results for duration and mean counts of pivoting. Finally, the investigation of the manually coded behaviour of rearing revealed no significant differences across genotype, dose and interactions.

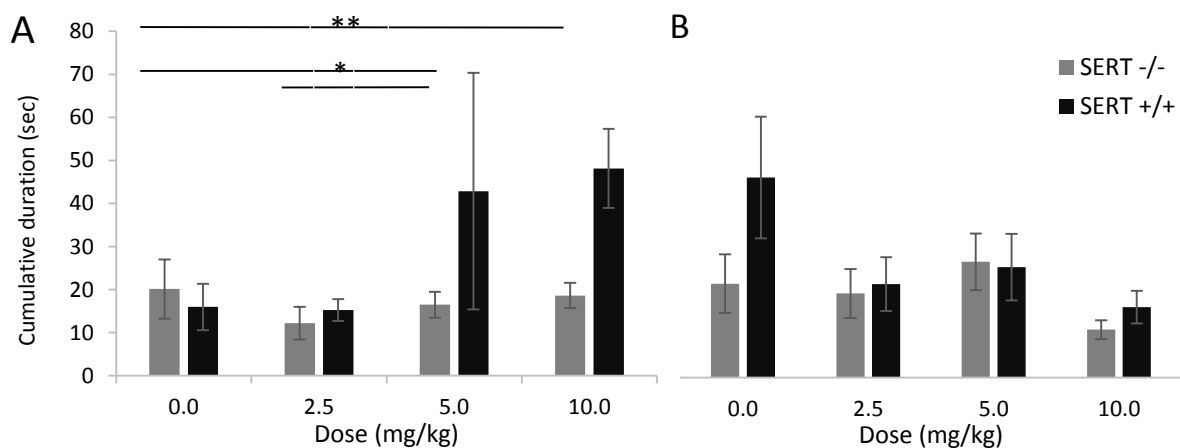


Figure 6. Mean duration of (a) pivoting and (b) rearing in SERT^{-/-} and SERT^{+/+} animals following administration of MDMA. Error bars represent standard error of the mean. * $p < .05$, ** $p < .01$.

Locomotor Activity following Amphetamine Administration

In a multivariate factorial analysis, the manually coded behaviour of grooming (mean count of grooming events in the first 30 minutes) revealed a significant main effect of genotype ($F(1,58) = 6.643$, $p < .05$, $\eta^2 = .103$) and a main effect of dose ($F(3,58) = 9.293$, $p < .001$, $\eta^2 = .319$). The analysis of total duration of grooming also revealed significant effect of dose, ($F(3,58) = 9.073$, $p < .001$, $\eta^2 = .325$). This dose effect was driven by the differences between 0.0mg/kg and the three other doses ($p < .05$; as per fig. 7).

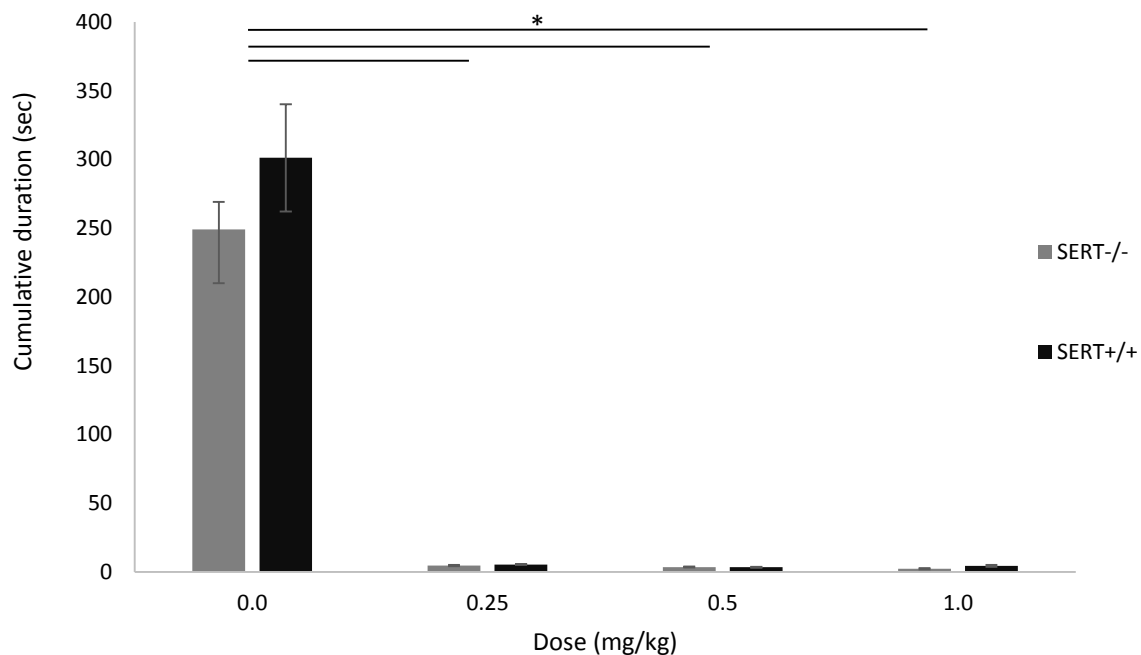


Figure 7. Mean duration of grooming behaviour, in SERT^{-/-} and SERT^{+/+} animals following administration of AMPH. Error bars represent standard error of the mean * $p < .05$

The manually coded behaviour of head weaving revealed a main effect of genotype ($F(1, 58) = 6.650$, $p < .05$, $\eta^2 = .103$) following AMPH administration SERT^{-/-} animals spend significantly more time head weaving than SERT^{+/+} animals. There was no effect dose however this relationship is approaching significance ($p = .052$ *n.s.*; as per fig. 8), suggesting that this behaviour is impacted other factors, like general activity. Also, no genotype and dose interactions were identified.

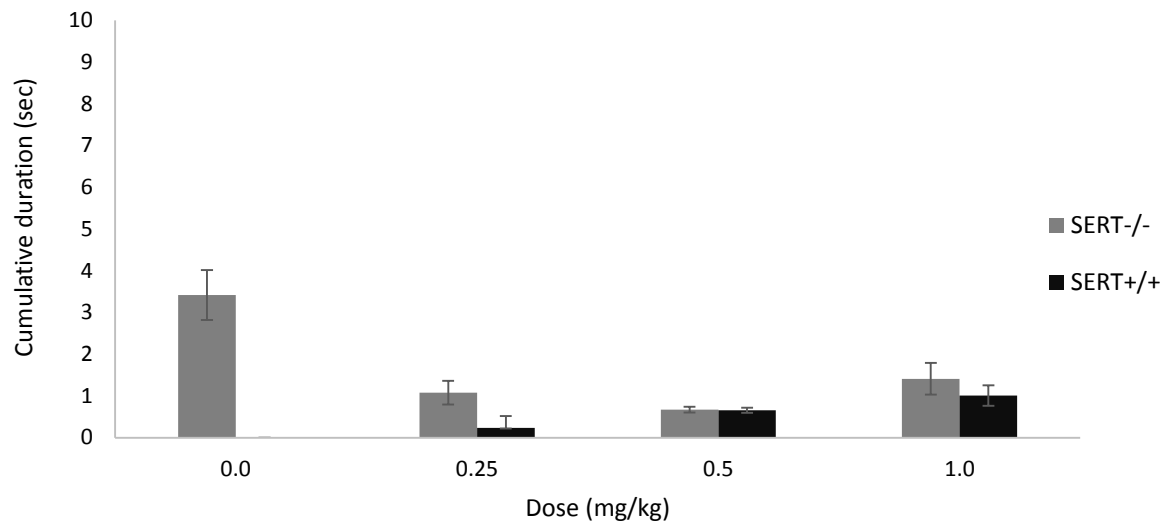


Figure 8. Mean duration of the head weaving behaviour, in SERT^{-/-} and SERT^{+/+} animals, following administration of AMPH. Error bars represent standard error of the mean.

The analysis of the pivot behaviour revealed a main effect of dose for the duration of pivoting, ($F(3, 58) = 6.613, p < .001, \eta^2 = .255$), this was driven by a significant difference between the dose of 0.0mg/kg and 0.5mg/kg ($p = .001$; as per fig. 9a). However, the genotype and interaction analyses revealed no other significant results for duration and mean counts of pivoting. The analysis of the rearing behaviour revealed a main effect of genotype ($F(1, 58) = 7.015, p < .05, \eta^2 = .108$) but no other dose or interaction effects were significant (as per fig. 9).

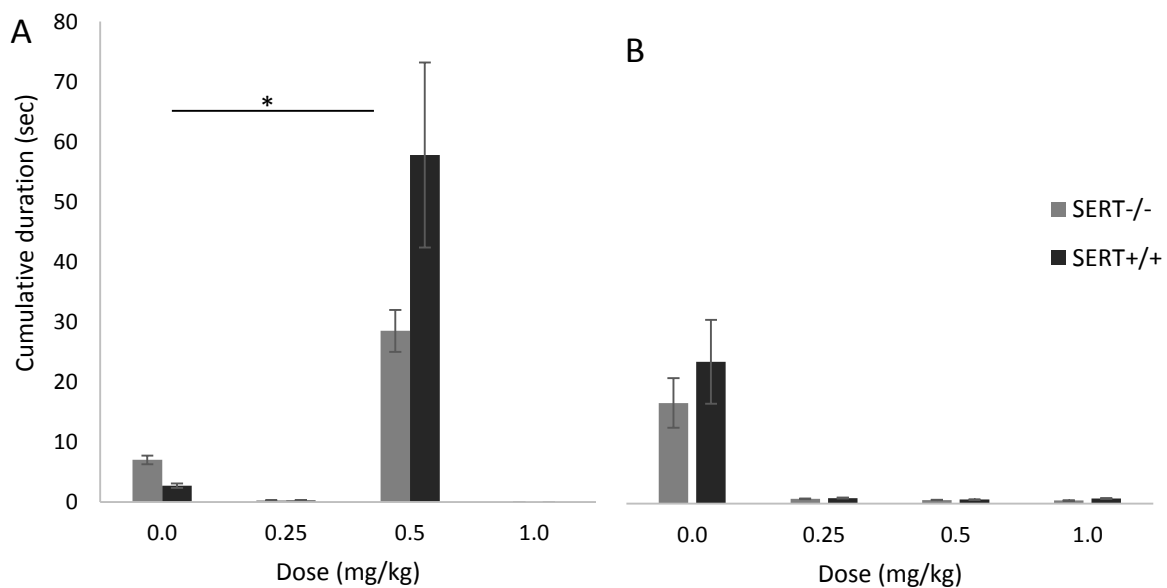


Figure 9. Mean duration of (a) pivoting and (b) rearing, in SERT^{-/-} and SERT^{+/+} animals following administration of AMPH. Error bars represented standard error of the mean. * $p < .05$.

Correlations

A series of Pearson's correlations were run to examine the relationship between the manually coded data and the data extracted from EthoVision XT (appendix one, table 1). In review of the Pearson's correlations, each of the manually coded behaviours was significantly correlated with at least one variable that was measured by EthoVision. Overall, these correlations demonstrate the relationship between both forms of measurement for the locomotor activity in animals.

Examining the Pearson's correlations in the data collected from the MDMA group, the manually coded behaviour of head weaving was found to have the least amount of significant correlations with the variables from EthoVision. In particular, head weaving had a weak positive correlation with the stretched body posture ($r = .312, p < .05$), but also had moderately negative correlations with the turn angle of the tail base ($r = -.246, p < .05$) and meandering of the center point ($r = -.287, p < .05$). However, the manually coded behaviour of grooming showed increased number and strength of correlations with more of the EthoVision variables. Of interest, grooming was significantly positively correlated with the measure of turn angle with a range of correlations from $r = .502 (p < .001)$ through to $r = .591 (p < .001)$. Additionally, the manually coded behaviour of pivoting showed the significant correlations, however most were moderately negative, with a range of correlation from $r = -.534 (p < .001)$ through to $r = .259 (p < .05)$.

Additionally, for the variables recorded there was a strong positive relationship between the data collected in the first 30 minutes of the trial and the second 30 minutes, thus suggesting that the data obtained from the first half hour can be extrapolated to the second half hour (see appendix two).

Analysis of data obtained from Ethovision for the full 60 minutes

MDMA induced LMA

Distance travelled and velocity, are two variables which measure the overall movement of animal. Distance travelled showed a main effect of genotype ($F(1, 65) = 9.884, p < .05, \eta^2 = .132$), showing that the distance travelled was significant between the two genotype groups of SERT^{-/-} and SERT^{+/+}. There was also a significant main effect of dose, ($F(3, 65) = 28.812, p = .001, \eta^2 = .571$).

The post hoc comparison showed that the distance travelled on the 0.0mg/kg was significantly different from 5.0mg/kg ($p = .001$) and 10mg/kg ($p = .001$). Furthermore, 2.5 mg/kg was significantly different from 5.0mg/kg ($p = .001$) and 10mg/kg ($p = .001$). Also there was a significant genotype by dose interaction, ($F(3, 65) = 3.601, p < .05, \eta^2 = .143$). As figure 10 shows, the effects of MDMA was stronger in the SERT^{+/+} than the SERT^{-/-} rats. The same was found for the velocity parameter (genotype: $F(1, 65) = 7.731, p < .05, \eta^2 = .106$; dose, $F(3, 65) = 38.924, p = .001, \eta^2 = .642$; interaction, $F(3, 65) = 4.065, p = .05, \eta^2 = .158$).

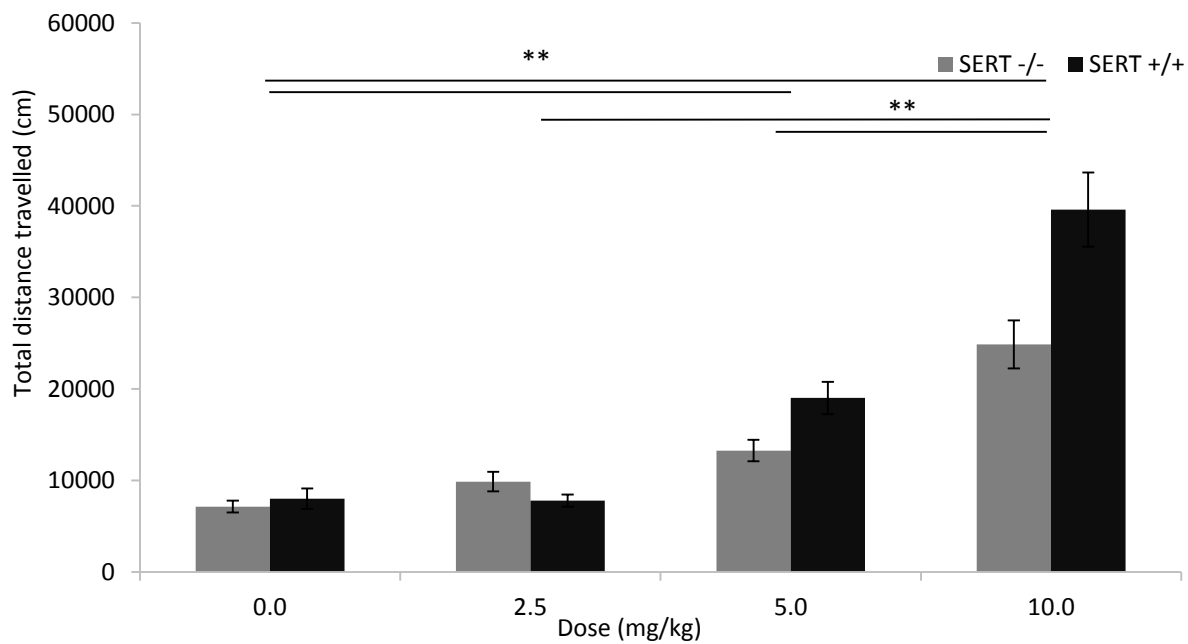


Figure 10. Means of distance travelled of SERT^{-/-} and SERT^{+/+} full trial, MDMA doses of 0.0, 2.5, 5.0, 10.0 mg/kg error bars represent the standard error. ** $p < .001$.

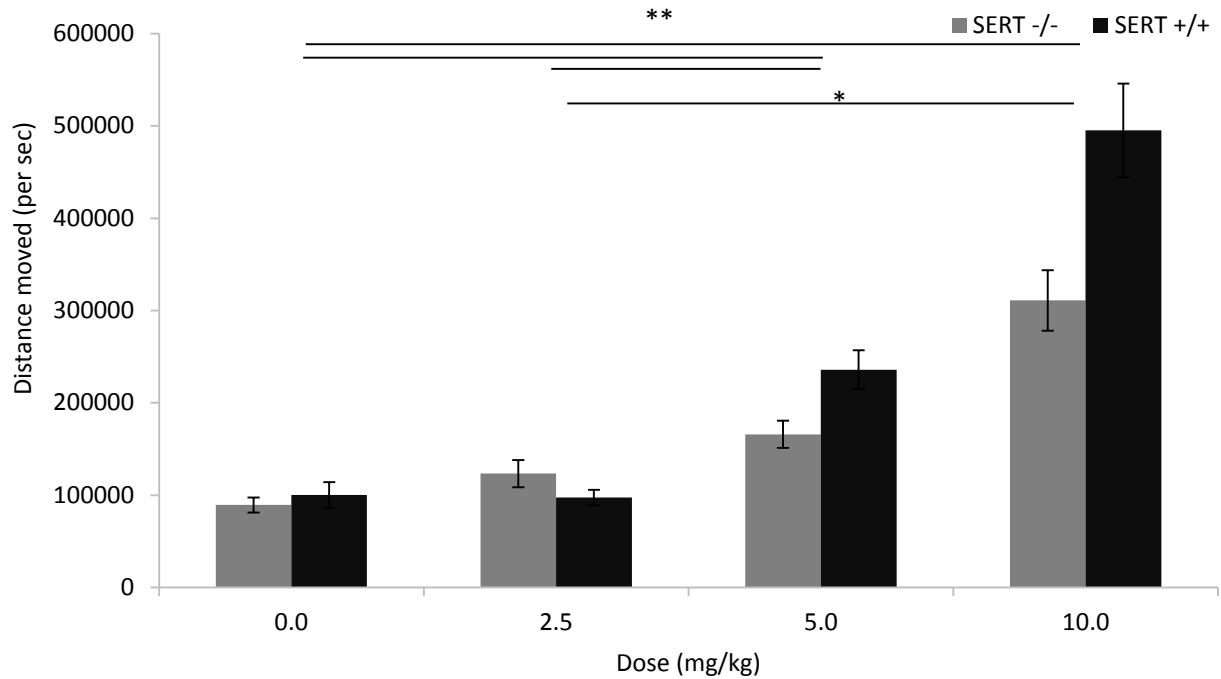


Figure 11. Means of velocity for SERT^{-/-} and SERT^{+/+} animals following MDMA administration. Error bars represent the standard error. * $p < .05$, ** $p < .001$.

Given the large number of variables obtained with EthoVision, in the present description we focus on those that are most relevant for our purpose and show significant genotype, dose and/or interactions. No genotype or dose effects or interactions were found with respect to contracted, normal, and stretched body posture.

One of the most interesting parameters is the turn angle, which is defined as the change direction of each body point and is measured in degrees (0°-180° per second). For the three body points there was a significant genotype effect (NP: $F(1, 65) = 5.720$, $p < 0.05$, $\eta^2 = .081$; CP: $F(1, 65) = 8.208$, $p < 0.05$, $\eta^2 = .112$; TB: $F(1, 65) = 7.069$, $p < 0.05$, $\eta^2 = .098$). Likewise there was a significant main effect of dose (NP: $F(3, 65) = 17.420$, $p = 0.001$, $\eta^2 = .446$; CP: $F(3, 65) = 19.974$, $p = 0.001$, $\eta^2 = .480$; TB: $F(3, 65) = 12.257$, $p = 0.001$, $\eta^2 = .361$) but no significant interaction. Post hoc analysis of the dose effects showed that the 0.0mg/kg was significantly different from 5.0mg/kg and 10.0mg/kg for all three body points.

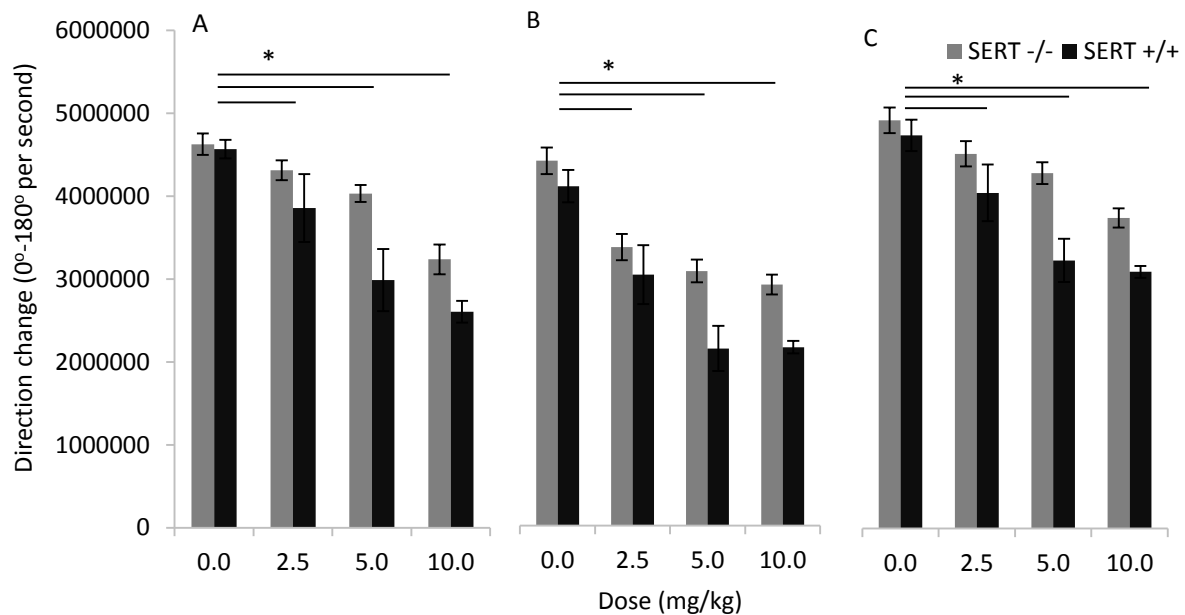


Figure 12. Means comparison of turn angle across NP(a), CP (b) and TB(c) , in SERT^{-/-} and SERT^{+/+} animals following MDMA administration. Error bar represent standard error of the mean. * $p < .05$.

The final group of variables to be evaluated is the measure of meandering, with individual measures of the nose, center and tail base points. For the analysis of meandering of the nose point, there was no main effect of genotype or dose. The analysis of meandering of the center point showed no main effect of genotype but there was a significant main effect of dose, ($F(3, 65) = 14.128$, $p = .001$, $\eta^2 = .395$) and the post hoc comparison showed that the doses of 0.0mg/kg and 2.5mg/kg were significantly different from 10.0mg/kg ($p = .001$ for both tests) and 5.0mg/kg was significantly different from 10mg/kg ($p < .05$), In the analysis of meandering of the tail base point, there was no main effect of genotype or dose. Overall, there was no genotype by dose interactions detected in the analysis of meandering for nose, center and tail base points.

AMPH induced LMA

For distance travelled following AMPH administration there was a main effect of genotype ($F(1, 58) = 9.408$, $p < .05$, $\eta^2 = .140$); and dose, ($F(3, 58) = 29.148$, $p = .001$, $\eta^2 = .605$) with the post hoc comparison showing that 0.0mg/kg was significantly different from 0.5mg/kg ($p = .001$) and 1.0mg/kg ($p = .001$). Furthermore, the measure of velocity showed the same effects, beginning with a main effect of genotype ($F(1, 58) = 17.123$, $p = .001$, $\eta^2 = .228$); dose, ($F(3, 58) = 29.572$, $p = .001$, $\eta^2 = .605$) with post hoc comparison again confirming that 0.0mg/kg was significantly different from

0.5mg/kg ($p = .001$) and 1.0mg/kg ($p = .001$). Finally, for the measure of velocity, there was a significant genotype by dose interaction ($F(3, 58) = 5.321, p < .05, \eta^2 = .216$). However, this interaction was opposite to the one seen with MDMA, with AMPH having a stronger effects in the SERT^{-/-} compared to the SERT^{+/+} rats.

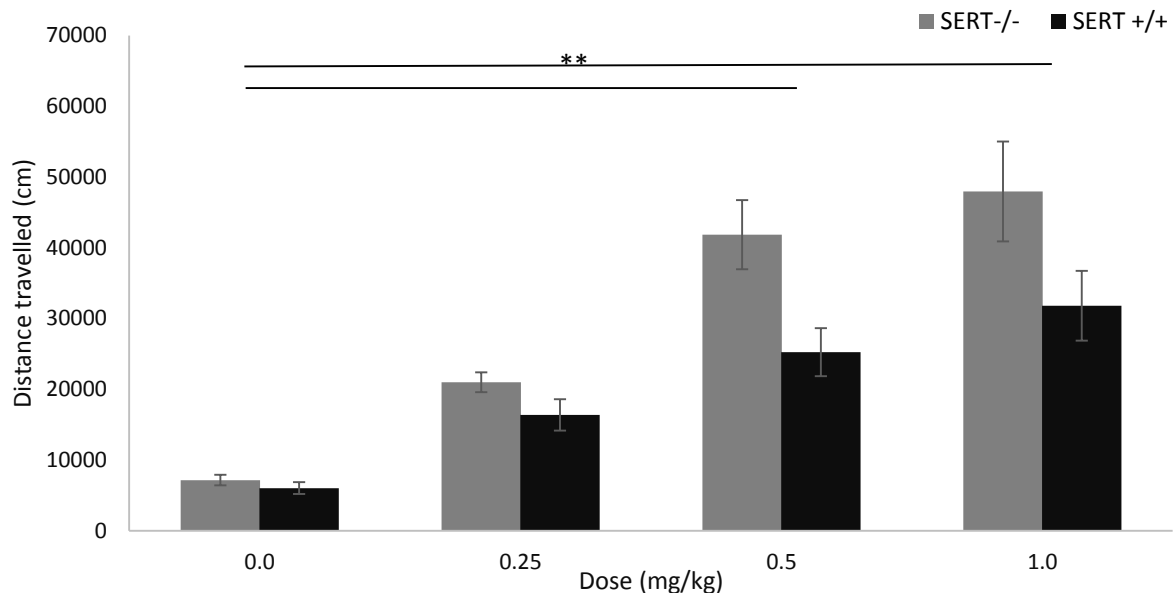


Figure 13. Means of distance travelled of SERT^{-/-} and SERT^{+/+} animals following AMPH administration. Error bars represent the standard error of mean. ** $p < .001$

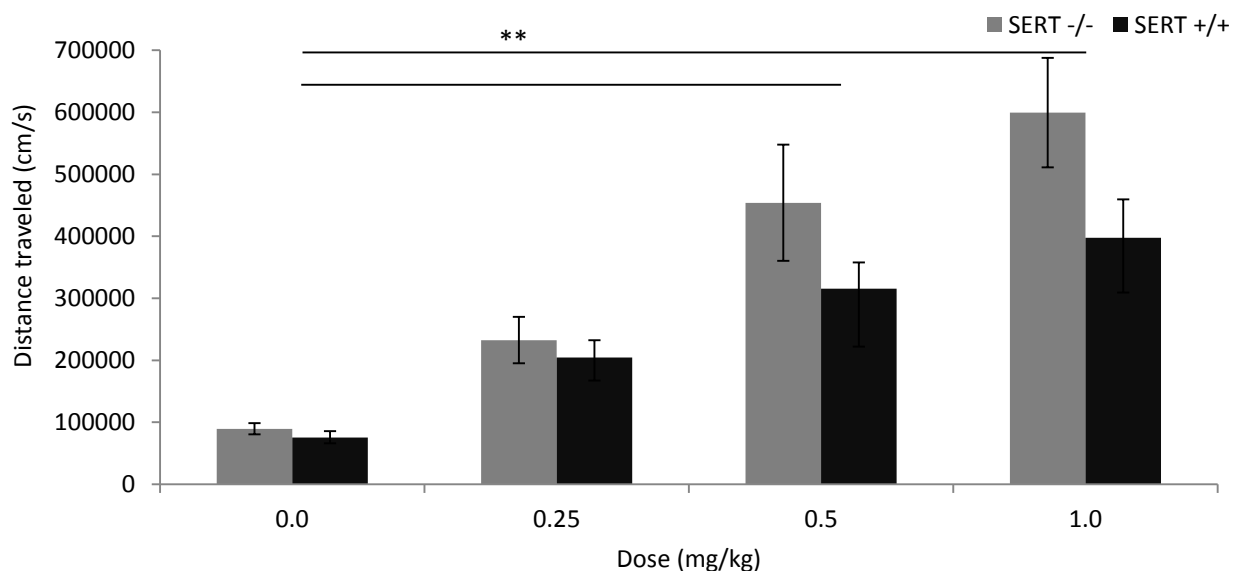


Figure 14. Means of velocity of SERT^{-/-} and SERT^{+/+} animals following AMPH administration. Error bars represent the standard error of mean. ** $p < .001$

There was a main effect of genotype ($F(1, 58) = 15.201, p = .001, \eta^2 = .208$) and dose ($F(3, 58) = 3.731, p < .05, \eta^2 = .162$) for the stretched body posture, but no interaction.

The examination of turn angle across nose, center and tail base revealed significant differences. While in this case there were no significant genotype effects, there were, again significant dose effects: (NP: $F(3, 58) = 22.829, p = .001, \eta^2 = .541$; CP $F(3, 58) = 45.162, p = .001, \eta^2 = .700$; TB: $F(3, 58) = 20.221, p = .001, \eta^2 = .511$). Post hoc analyses showed that the 0.0mg/kg dose was significantly different from all other doses of amphetamine (as per figure 15).

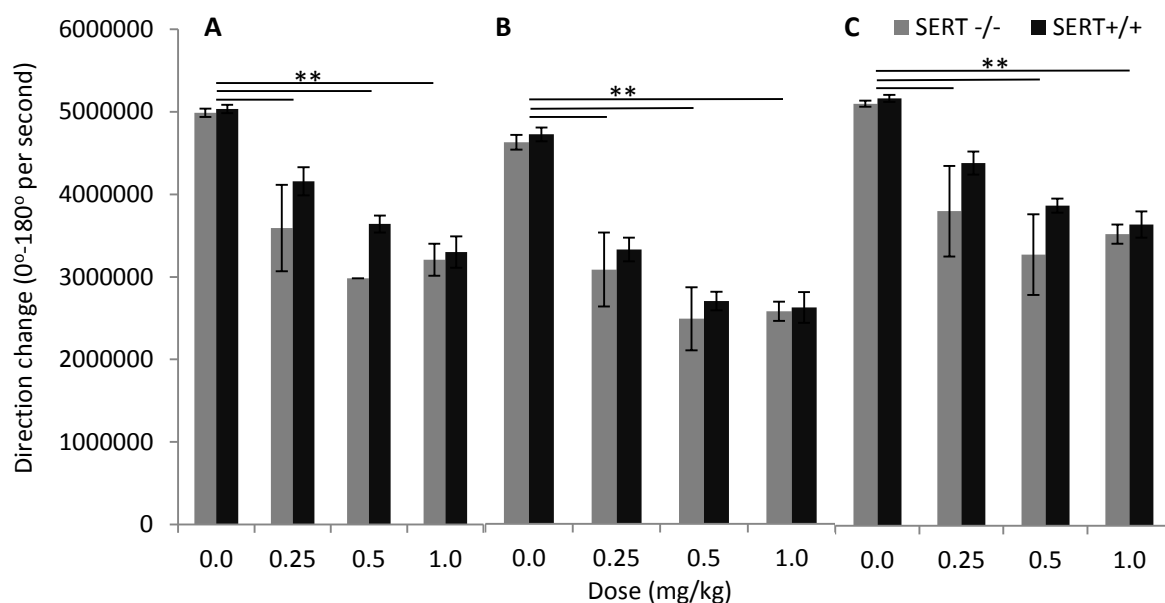


Figure 15. Means comparison of turn angle across NP(a), CP (b) and TB(c) , in SERT^{-/-} and SERT^{+/+} animals following AMPH administration. Error bars represent standard error of the mean. * $p < .05$.

Finally, in the analysis of meandering, there was a significant main effect of dose for the center point of the animal ($F(3, 58) = 80.067, p = .001, \eta^2 = .806$) and the post hoc comparison showed that 0.0mg/kg was significantly different from 0.25mg/kg, 0.5mg/kg and 1.0mg/kg ($p = .001$ for all comparisons) and 2.5 mg/kg was significantly different from 1.0mg/kg ($p = .001$). In the analysis of meandering of the nose point and tail base point, there were no main effects of genotype or dose. Overall, there was no genotype by dose interactions detected in the analysis of meandering for nose, center and tail base points.

Discussion

The aim of the current study was to distinguish the behavioural patterns of AMPH and MDMA using the SERT^{-/-} animal which is a genetic model of SERT deletion and provides an avenue for investigating the role of 5HT in behaviour. While the rewarding effects of many drugs of abuse have been attributed to dopaminergic (DA-ergic) systems, there are other behavioural patterns driven by other neural mechanisms, such as the serotonergic (5HT-ergic) systems. The administration of MDMA or AMPH result in hyperactivity, increases in heart rate and blood pressure and stereotyped behaviour such as sniffing, grooming, pivoting, head weaving and rearing. While both substances are classed as stimulants, the primary pharmacological effects are different. Whereas AMPH preferentially releases DA, MDMA acts as a 5HT releasing agent. In order to measure the unique behaviour profiles of these two drugs, and the relevant contribution of 5HT and DA, animals lacking the SERT protein were used.

It is through the use of the SERT^{-/-} animal model that it is possible to dissociate the 5HTergic and DA-ergic effects of MDMA and AMPH. As these animals lack the SERT protein, when MDMA is administered it cannot act upon the protein and as such the behavioural outcomes observed in this animal are dominated by DA-ergic actions. The 5HTergic effects following MDMA treatment and resulting excess of 5HT in the neuronal environment is the source of the SS. In rats this results in displays of head weaving, fore-paw treading, and salivation (Spanos & Yamamoto, 1989) and hyperactivity (Gold & Koob, 1989). However, hyperactivity is also a behavioural outcome following AMPH treatment which is predominately a DA releaser. While this relationship was acknowledged by Green, et al. (2003), there is lack of evidence for the distinguishing facets of hyperactivity from MDMA treatment compared with AMPH. As such the SERT^{-/-} animal model provides a novel method for defining the distinctions of hyperactivity induced by MDMA and that which is induced by AMPH.

Using the drug discrimination paradigm, it was possible to measure possible interoceptive cues experienced by SERT^{-/-} and SERT^{+/+} animals. This paradigm requires the animals to identify if a challenge drug (AMPH) *feels* more like the drug they were trained on (MDMA) or a vehicle solution. If an animal lacking the SERT does not experience 5HT release following acute MDMA

administration, then it can be assumed that any effects are due to the DA contribution suggesting that AMPH and MDMA would be indistinguishable resulting in generalization of the cue. However, for animals with functioning SERT, that can experience 5HT effects of MDMA and DA effects of AMPH, it would be expected that 0.5mg/kg AMPH would not fully generalize to MDMA, that is, it would *feel* similar but not the same as MDMA. Additionally, as the DA-ergic contribution following 1.5mg/kg MDMA is much weaker than 5HT, it was expected that the SERT^{+/+} will acquire drug discrimination faster than SERT^{-/-}.

During drug discrimination, animals were trained to discriminate between a low dose of MDMA and vehicle. If an animal responded more than 85% on the correct drug-paired lever during the first FR10 during three consecutive sessions, they were considered ‘acquired’. Not only did more SERT^{+/+} animals reach acquisition, they reached acquisition twice as fast as the SERT^{-/-} animals. This supports the idea that the stimulus effects of MDMA were stronger for the SERT^{+/+} animals, as the contribution of MDMA released 5HT is greater in these animals. This finding aligns with previous research which had shown that rats were able to discriminate between MDMA and the vehicle (Oberlander & Nichols, 1988) and expands on previous literature demonstrating the salience of the 5HT-ergic cues of MDMA for the SERT^{+/+} animals.

Once animals had reached acquisition, a generalisation test was conducted using 0.5mg/kg AMPH in order to measure whether animals can distinguish between the pharmacological effects of MDMA and vehicle. Although less SERT^{-/-} animals reached this phase of the experiment, performance during the generalisation tests was enhanced, as shown by a significant decrease in the time taken to reach the initial FR10. This suggests that the SERT^{-/-} animals are more sensitive to the stimulus effects of AMPH compared to the SERT^{+/+} animals. This further supports the idea that SERT^{-/-} animals will completely generalise AMPH to MDMA, due to the primary effect on DA-ergic systems by both drugs. In contrast, SERT^{+/+} animals were not expected to fully generalise AMPH to MDMA, as MDMA results in both 5-HTergic and DA-ergic effects for these animals compared to the sole DA-ergic effects of MDMA for the SERT^{-/-} animals.

Accordingly, it could be expected that SERT^{+/+} animals would partially generalise the MDMA paired lever with AMPH but that this preference would not be as great as for the SERT^{-/-} animals. While the SERT^{+/+} animals showed no lever preference at all, there was a slight preference for the MDMA lever by the animals, however this was non-significant. This may be due to the large amount of variation seen and the low sample size (6 and 4 respectively) used in the current study. While some animals completed four generalisation tests, the extended amount of time needed for SERT^{-/-} animals to acquire meant that some of these animals only completed one generalisation test.

The drug discrimination data has demonstrated that the two types of animals' experience different stimulus properties from MDMA. Both the number of SERT^{+/+} animals acquiring and the decreased latency show that discrimination of related cues develops differently in these animals. Whereas, for the SERT^{-/-} animals the salience of the stimulus effects of AMPH was heightened, demonstrated by the lower response time during the first FR10. However, at the training dose of 1.5mg/kg and the generalisation dose of 0.5mg/kg did not lead to complete generalisation in the lever preference, and this may be related to other stimulus properties which may underlie the effects of MDMA and AMPH, such as NE. An alternative explanation is that perhaps the training is too low, preliminary work by Harper, Webster and Schenk (unpublished) has shown that animals trained on a higher dose of MDMA reliably respond on the MDMA-paired lever during generalisation tests with AMPH. Additionally, drawing from the findings in the LMA experiment of this study, the differences between 0.0mg/kg and 2.5mg/kg of MDMA or 0.0mg/kg and 0.5mg/kg of AMPH is often not significantly different. While the LMA drug administrations are not comparable to drug discrimination, it does allude to the possibility that lever preference might be present if the training doses are increased.

Using the locomotor activity paradigm, it was possible to measure the behavioural patterns of MDMA and AMPH in SERT^{-/-} and SERT^{+/+} animals, with a particular focus on the hyperactivity induced by these stimulants. Multiple measures were reported to provide a cohesive picture of the behavioural pattern stimulated. The gross hyperactivity profile for the complete trial was provided by automatic tracking using EthoVision, while the more complex behaviours were manually coded for

the first 30 minutes of the trial. The data obtained from both measures was correlated to examine the relationship between manually coded behaviours and video tracked, as well as the relationship between the first and second halves of the trials.

As previously discussed, the 5HTergic effects following MDMA administration do not impact the behaviour of the SERT^{-/-} animals and as such it was expected that these animals would have reduced behavioural displays grooming, head weaving, pivoting and rearing when compared to SERT^{+/+}. However there were no identifiable differences across SERT^{-/-} and SERT^{+/+} animals for these behaviours following MDMA administration. While, Spanos and Yamamoto (1989) as well as Antonious, Kafetzopoulos, Papadopoulou-Daifoti, Hyphantis and Marselos (1998) had found manually coding behaviours to be a valuable method, the results of this study have not aligned. The lack of identified difference may be attributed to the lack of sensitivity of the manually coded process, whereby the quantitative differences between SERT^{-/-} and SERT^{+/+} animals was not statistically distinguishable when collected through this manual process for this group of animals.

Additionally, the manually coded behaviours of grooming, head weaving, pivoting and rearing in SERT^{-/-} and SERT^{+/+} animals following AMPH administration was examined. The analyses showed a significant genotype difference in the grooming behaviour for SERT^{-/-} and SERT^{+/+} animals and this effect was present for head weaving and rearing. These analyses demonstrate that SERT^{-/-} and SERT^{+/+} animals are effected differently by AMPH and perhaps these behaviours are closely aligned with hyperactivity following AMPH administration rather than MDMA administration.

In addition to the manually coded behaviours, this study used the video-tracking system EthoVision, as this allows for a finer-grained analysis of the animals body points across multiple measures. For the purposes of continuity and reliability of the measures, correlations were used to examine the relationships between the data extracted from manually coding the animal's behaviour and the data extracted from EthoVision tracking. The analyses of these correlations revealed that the manually coded behaviours were related to at least one variable measured by EthoVision. Thus, both methods of measuring providing an avenue for examining animal behaviours. As the manually coded

data was from the first 30 minutes of the trial, it was also of value to investigate the relationship between the first and second 30 minutes of the trial. Overall, this analysis showed a strong positive relationship between the two time frames. As such this demonstrates that the manually scored behaviours taken during the first 30 minutes are a good indication of the behaviours across the full trial and that the data extracted from the full trial is a fair representation of the animal's behaviour.

In light of the relationships between data extracted from manually coded and the data extracted from EthoVision, it is possible focus on measures of hyperactivity following the administration of MDMA and AMPH in SERT^{-/-} and SERT^{+/+} animals. For this study, it was expected that the SERT^{-/-} animals would have decreased levels of hyperactivity as the current literature indicates that the MDMA induced hyperactivity is mediated by 5HT-ergic effects of the drug. To examine behaviour measures focus on distance travelled and velocity as an approximation of general activity, as well as turn angle which is a measure of change in direction, and there are additional measures of body posture and meandering.

As a measure of the distance travelled by the animals during the full trial, there were differences between SERT^{-/-} and SERT^{+/+} animals, such that following the MDMA treatment SERT^{+/+} animals travelled more than in SERT^{-/-} animals. This was distinctive characterised on by the SERT^{+/+} animals travelling more when on doses of 5.0mg/kg and 10.0mg/kg of MDMA. The same effect was found in the measure of velocity, whereby SERT^{+/+} animal's motion was increased on doses of 5.0mg/kg and 10.0mg/kg. As a general measure of hyperactivity, these measures have demonstrated differences in the response of SERT^{-/-} and SERT^{+/+} animals to MDMA. At a broad level, this aligns with Gold and Koob (1989) and Green, et al. (2003) which demonstrated that hyperactivity is a definable characteristic of MDMA administration in rats. However, this finding demonstrates a pattern of hyperactivity following MDMA administration in SERT^{-/-} animal which do have the SERT protein for the 5HTergic effects to act upon. As such the SERT^{-/-} animals do show dose-dependent increases in the distance travelled and velocity of movement, but those increases are lower than SERT^{+/+} animals which showed the higher rates of activity. Overall, this suggests that the DA-ergic effects of MDMA contribute in part to the increases in hyperactivity.

In addition to the distance travelled and velocity, another measure to consider as part of the increased LMA induced by MDMA is turn angle which is the change in direction of the body points throughout the hours. This analysis showed that there were differences in the overall change in direction for the SERT^{-/-} and SERT^{+/+} animals. These differences are distinctively clear in the doses of 5.0mg/kg and 10.0mg/kg of MDMA, such that SERT^{-/-} animals show a decrease the direction change when compared with SERT^{+/+} on the same doses. This indicates that there are distinctive differences in the patterns of behaviour for the SERT^{-/-} and SERT^{+/+} animals, particularly with regard to the directional changes made during the trial and on the doses of 2.5mg/kg and 5.0mg/kg.

In the interests of capturing different behaviours, the additional measures of body posture and meandering were examined for differences in the LMA induced by MDMA. The analysis of the contracted, normal and stretched body posture frequencies, revealed no genotype effects; this was also the case for the measure of meandering. The lack of identified significance is perhaps related to large amounts of variability involved with such fine-grained analyses and it does provide ground for further investigation.

Pijnenburg, et al. (1976) and Antonious (1998) had shown that the treatment with AMPH results in LMA increases which are predominately driven by DA-ergic effects. In this study, it was predicted that the SERT^{-/-} animals would have locomotor activity changes induced by AMPH which were comparable to the SERT^{+/+} animals. The tests for this prediction revealed a difference in the distance travelled and overall rate of movement between SERT^{-/-} and SERT^{+/+} animals as well as there was the addition of a gene by dose interaction for velocity. This revealed that the SERT^{-/-} animals travelled more and at an increase rate of movement, than the SERT^{+/+} which is the reversed finding when compared to what was found following MDMA administration. This suggests that the functionally different behavioural profiles between the SERT^{-/-} and SERT^{+/+} animals following the acute treatment of AMPH. Furthermore the SERT^{-/-} animals are more sensitive to the stimulus effects of AMPH when compared to SERT^{+/+}, demonstrated by increased distance and velocity.

Another measure used to investigate the changes in LMA following the administration of AMPH is turn angle. The analyses of this measure revealed no significant differences between the SERT^{-/-} and SERT^{+/+} animals which is perhaps attributed to the variance involved, particularly in the SERT^{-/-} animals on doses of 0.25mg/kg and 0.5mg/kg of AMPH. Furthermore, the analyses of body posture revealed a difference in the total duration spent in a stretched posture between the SERT^{-/-} and SERT^{+/+} animals; which supports current finding of the differential effects of AMPH on these animals. Furthermore, the analysis of the animal's path with the use of meandering, identified no differences between the SERT^{-/-} and SERT^{+/+} animals. Potentially, these findings indicate that the more detailed behaviours of SERT^{-/-} and SERT^{+/+} animals are comparably effected by AMPH administration. However, given the previous finding that SERT^{-/-} animals travel more and at an increased rate follow AMPH administration, it is likely to be the case that turn angle, body posture and meandering are not clear indicators of the behaviours which are differentially effected by AMPH.

The alterations in the behaviours of SERT^{-/-} and SERT^{+/+} animals changes according to the doses of MDMA and AMPH. In this study it was predicted that there would be dose-dependent increases of behaviour for SERT^{-/-} and SERT^{+/+}. Overall, the analyses revealed significant differences typically with the behaviour at a dose of 0.0mg/kg when compared to 5.0mg/kg or 10.0mg/kg of MDMA, or the difference between 0.0mg/kg compared to 0.5mg/kg and 1.0mg/kg, and at times there was differences relating to the 2.5mg/kg of MDMA or 0.25mg/kg of AMPH. These dose dependent differences provide evidence for the change in stimulus cue with the increases of dose. Particularly within the MDMA doses, 0.0mg/kg and 2.5mg/kg appear comparable which often results in a lack of statistical difference between the behaviours on these doses. As such the higher doses of 5.0mg/kg and 10.0mg/kg of MDMA and 0.5mg/kg and 1.0mg/kg of AMPH are required to demonstrate and distinguish the behavioural differences relating to the stimulus cues of MDMA or AMPH.

Overall, the fine-grained analyses of LMA in SERT^{-/-} and SERT^{+/+} animals has demonstrated fundamental differences in behaviour after MDMA and AMPH administration. Such that the SERT^{-/-} animals display reduced levels of activity following MDMA administration, whereas the activity

levels of these animals is increased following AMPH administration. As such, the SERT^{-/-} are more sensitive to AMPH but the definable behaviours, a part from distance moved, is yet to be uncovered. However, these findings are perhaps compounded by the lack of within subject comparison. Whereby, the animals used to examine MDMA induced LMA were not the same animals as those used for the AMPH induced LMA. As such there may have been fundamental differences such as age and previous handling that may have contributed to overall behaviour of the animals and due to time constraints on this project and ethical restrictions these factors could not be controlled. Perhaps a future study could use the same animals for MDMA and AMPH induced LMA but use longer washout period between doses to inhibit previous drug effects on the behaviour. Furthermore, clearer definitions on behaviour may assist in the pinpointing the differential behaviours of MDMA and AMPH administration in the SERT^{-/-} and SERT^{+/+} animals. While this study has taken a fine-grained analysis of the drug induced LMA, the behaviours could have been refined further. Due to time constraints on this project, this finer-grained analysis was not completed.

In summary, this study aimed to distinguish the behavioural patterns of AMPH and MDMA using the SERT^{-/-} animal to examine the role of 5HT in behaviour when compared with SERT^{+/+} animals. The behavioural patterns was examined with drug discrimination and locomotor activity paradigms. The findings from the drug discrimination project demonstrate that the SERT^{-/-} animal is differently effected by MDMA and AMPH, such that these animals acquired discrimination later but then showed enhanced response time during the generalisation test, although there was no preference for drug-paired lever. These findings demonstrate that the salience of the stimulus effects of AMPH was heightened for the SERT^{-/-} animal, however there may be related stimulus properties which may underlie the effects of MDMA and AMPH, such as NE. Furthermore, distinctive behavioural differences was identified in the LMA paradigm with the SERT^{-/-} animal displaying inhibited levels of hyperactivity following MDMA administration but then hyperactivity was increased following AMPH administration. In conclusion, the behavioural pattern induced by MDMA administration in the SERT^{-/-} animal is inhibited, demonstrated by reductions in acquisition and hyperactivity.

However, the pattern induced by AMPH administration in the SERT^{-/-} animal is enhanced, as seen with quicker responses during generalisation trials as well as the measures of hyperactivity.

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Appendix One

EthoVision Variables	Manually coded behaviours			
	Grooming	Head weaving	Pivoting	Rearing
Distance travelled time 1	-.479 **			.543**
Distance travelled time 2	-.517**		.259*	.475**
Contracted body posture time 1	.591**			
Contracted body posture time 2				
Normal Body posture time 1			-.369**	
Normal body posture time 2			-.290*	
Stretched body posture time 1				
Stretched body posture time2		.312*		
Turn angle of the nose time 1	.537**			-.554**
Turn angle of the nose time 2	.591**		-.380**	-.424**
Turn angle of center time 1	.556**		-.428**	-.485**
Turn angle of center time 2	.568**		-.425**	-.484**
Turn angle of tail base time 1	.502**		-.363**	-.369**
Turn angle of tail base time 2	.545**	-.246*	-.478**	
Meandering of the nose time 1			-.366**	
Meandering of the nose time 2			-.473**	
Meandering of the center time 1	.507**		-.424**	-.412**
Meandering of center time 2	.587**	-.287*	-.398**	-.338**
Meandering of the tail base time 1				
Meandering of the tail base time 2			-.534**	.594**

Table 1 Correlations of Ethovision variables and manually coded behaviours with data extracted from the MDMA data set. Time 1: 0:00 – 30:00 mins, time 2: 31:00 – 60:00 mins * $p < .05$, ** $p < .01$.

EthoVision Variables	Manually coded behaviours			
	Grooming	Head weaving	Pivoting	Rearing
Distance travelled time 1	-.509**	.267*	.360**	
Distance travelled time 2		.311**		
Contracted body posture time 1	.339*			
Contracted body posture time 2	.316*			
Normal Body posture time 1	-.313*			
Normal body posture time 2				
Stretched body posture time 1	-.238*			
Stretched body posture time2				
Turn angle of the nose time 1	.616**	-.426**	-.561**	
Turn angle of the nose time 2	.653**	-.284*	-.612**	
Turn angle of center time 1	.540**		-.547**	
Turn angle of center time 2	.578--		-.303*	
Turn angle of tail base time 1	.563**	-.324*	-.528**	
Turn angle of tail base time 2	.619**	-.310*	-.562**	
Meandering of the nose time 1				
Meandering of the nose time 2		-.356**		
Meandering of the center time 1	.368**			-.278*
Meandering of center time 2	.543**		-.266*	
Meandering of the tail base time 1				
Meandering of the tail base time 2	.421**			

Table 2 Correlations of Ethovision variables and manually coded behaviours with data extracted from the AMPH data set. Time 1: 0:00 – 30:00 mins, time 2: 31:00 – 60:00 mins * $p < .05$, ** $p < .01$.

Appendix Two

MDMA Variables	Time 1 – 0:00-30:00		Time 2 – 31:00-60:00		Pearson's Correlation (<i>r</i>)
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	Time 1 correlated to time 2
Distance travelled	8071.01	5393.98	9055.55	8650.11	.822**
Velocity	100887.62	67424.73	113194.27	108126.38	.822**
Contracted body posture	100.75	133.87	108.53	240.76	.812**
Normal Body posture	1694.85	133.04	1684.41	243.73	.811**
Stretched body posture	3.91	8.78	2.97	9.33	.259*
Turn angle of the nose	2.97	9.33	1930382.60	336624.61	.843**
Turn angle of center	1930382.60	336624.61	1950021.81	495069.92	.840**
Turn angle of tail base	1950021.81	495069.92	1529065.08	347000.28	.818**
Meandering of the nose	1529065.08	347000.28	1571354.03	520510.36	-.010
Meandering of the center	1571354.03	520510.36	2094808.77	285668.90	.202
Meandering of the tail base	2094808.77	285668.90	2102973.15	437918.58	-.707

Table 3 Pearson's correlation of variables measured in the MDMA data set across time 1 and 2, with means and SD. * $p < .05$, ** $p < 0.001$.

AMPH Variables	Time 1 – 0:00-30:00		Time 2 – 31:00- 60:00		Pearson's Correlations (<i>r</i>)
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	Time 1 correlated with time 2
Distance travelled	12058.67	8537.42	11842.32	9848.50	.822**
Velocity	11842.32	9848.50	150733.36	106717.81	.822**
Contracted body posture	150733.36	106717.81	148028.99	123106.18	.812**
Normal Body posture	148028.99	123106.18	47.10	79.35	.811**
Stretched body posture	47.10	79.35	71.18	186.73	.259*
Turn angle of the nose	71.18	186.73	1626.13	420.56	.843**
Turn angle of center	1626.13	420.56	1598.37	453.74	.840**
Turn angle of tail base	1598.37	453.74	19.65	28.87	.818**
Meandering of the nose	19.65	28.87	17.48	38.15	-.010
Meandering of the center	17.48	38.15	1906783.09	478491.38	.808**
Meandering of the tail base	1906783.09	478491.38	2025621.82	476553.97	-.070

Table 4 Pearson's correlation of variables measured in the AMPH data set across time 1 and 2, with means and SD. * $p < .05$, ** $p < .01$.

Appendix Three

Behaviours that were manually coded by the experimenters for the first 30 minutes of the trial are:

- Grooming
- Head weaving
- Pivot
- Rearing

Cumulative duration and the mean frequency of these behaviours was used for analysis in this study.

Measures obtained from EthoVision XT for the full 60 minutes of the trial are:

- Distance travelled (cm) of the animal's centre point
- Velocity of the animal's centre point
- Contracted body posture
- Normal body posture
- Stretched body posture
- Meandering of the nose point
- Meandering of the centre point
- Meandering of the tail base
- Turn angle of the nose point
- Turn angle of the centre point
- Turn angle of the tail base

The total was the measure extracted from these variables and used for analysis in this study. Total refers to the total distance travelled (cm) or the total frequency of the stretched body posture, or the total turn angle of the nose point (°-degrees).